Hybrid Quantum-Classical Framework for Multi-objective Weighted Portfolio Optimization

Yao-Hsin Chou
Dept. Computer Science and Information Engineering
National Chi Nan University
Puli, Taiwan
yhchou@ncnu.edu.tw

Kun-Lin Lee
Dept. Computer Science and Information Engineering
National Chi Nan University
Puli, Taiwan
s111321015@mail1.ncnu.edu.tw

Ping-I Lin
Dept. Computer Science and Information Engineering
National Chi Nan University
Puli, Taiwan
s111321024@mail1.ncnu.edu.tw

Yen-Yu Hung
Dept. Computer Science and Information Engineering
National Chi Nan University
Puli, Taiwan
s111352004@mail1.ncnu.edu.tw

Shu-Yu Kuo
Dept. Electrical Engineering
National Taiwan University
Taipei, Taiwan
shuyu@ntu.edu.tw

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***Abstract*—Quantum and quantum-inspired computations have shown remarkable potential in addressing complex optimization problems. Quantum annealing (QA), in particular, has demonstrated effectiveness in solving combinatorial optimization tasks and is gaining increasing attention. Meanwhile, quantum-inspired computing (QIC) algorithms offer a promising approach to leveraging quantum principles on classical hardware. This study presents the first integrated hybrid framework that combines the strengths of both quantum and classical computing for multi-objective weighted portfolio optimization, a complex real-world application. In the proposed approach, QA employs scalarization techniques to rapidly approximate the optimal front, while QIC enhances solution diversity through an entanglement move mechanism that uncovers a broader set of non-dominated portfolio solutions. Experimental results indicate that the hybrid approach achieves higher-quality solutions and provides a greater variety of investment options for investors, compared to QA alone and classical multi-objective optimization algorithms. These findings highlight the feasibility and scalability of integrating quantum and classical techniques to effectively address real-world investment optimization challenges.**

***Index Terms*—Quantum Computing, Quantum-inspired Computing, Multi-objective Weighted Portfolio Optimization**

I. Introduction

Quantum computing, including quantum annealing (QA), leverages quantum phenomena to enable faster and more efficient solutions to complex optimization problems. Concurrently, quantum-inspired computing (QIC) [1] draws on quantum principles to design algorithms that effectively solve such problems on classical hardware. Collectively, these approaches offer promising opportunities for addressing real-world optimization challenges.

Portfolio optimization [2] is a critical real-world problem in finance, inherently involving conflicting objectives. Some investors seek higher returns at the expense of increased risk, while others prioritize lower risk even with reduced potential gains. These opposing goals make portfolio optimization a natural application for multi-objective optimization (MOO), which explores trade-offs and generates a Pareto front to provide diverse, high-quality solutions tailored to varying investment preferences.

This study develops a hybrid MOO algorithm that combines the advantages of quantum and classical computing to address complex financial problems. The proposed method integrates a QA scalarization-based approach with a trend ratio model [3], further enhanced by a quantum-inspired entanglement move (EM). This combination enables the generation of diverse non-dominated portfolios that reflect a broad range of investment preferences. Experimental results on real-world financial data demonstrate promising performance compared to classical MOO algorithms, confirming the feasibility and competitiveness of the hybrid approach.

II. Methodology

This paper presents a novel hybrid model that integrates QA with QIC, bridging quantum and classical approaches.

A. Multi-objective Weighted Portfolio Model

The multi-objective weighted portfolio optimization model seeks to accommodate investors with different risk preferences, while simultaneously maximizing return and minimizing risk. Integer-valued weights are assigned to each stock to evaluate the daily expected return and daily risk of each portfolio under the trend ratio model. The resulting solutions form a non-dominated set (NDS) according to Equation (2), where a portfolio P_i is said to dominate P_j if $P_i \succ P_j$. All solutions within the NDS are non-dominated, representing optimal trade-offs between risk and return.

$$\begin{aligned} &Optimal\ f(P) = \{f_1(P),\ f_2(P)\},\ P = \{x_1, x_2, ..., x_N\} \\ &min\ f_1(P) : Daily\ Risk\ (DR) \\ &max\ f_2(P) : Daily\ Expected\ Return\ (DER) \\ &subject\ to\ \sum_{i=1}^{N} x_i \leq 100,\ x_i \in \mathbb{Z},\ 0 \leq x_i \leq 100 \end{aligned} \tag{1}$$

$$\begin{aligned} P_i \succ P_j \iff &(f_1(P_i) \leq f_1(P_j) \wedge f_2(P_i) \geq f_2(P_j)) \\ &\wedge (f_1(P_i) < f_1(P_j) \vee f_2(P_i) > f_2(P_j)) \end{aligned} \tag{2}$$

B. Multi-objective Quantum Annealing (MoQA)

The MoQA approach is shown in Equation (3). Under the constraints of the QA framework, this objective function is minimized by treating DR as a positive term and DER as a negative term. A scalarization weight parameter λ is used to control the ratio between the two objectives, generating solutions that reflect different preferences.

$$E(\omega) = \lambda \cdot DR - (1 - \lambda) \cdot DER \tag{3}$$

The value of λ lies within the interval $[0, 1]$, each λ corresponds to a distinct subproblem in the optimization process. When $\lambda = 0$, the contribution of DR is nullified, resulting in solutions that focus on maximizing return, regardless of risk. Conversely, when $\lambda = 1$, the contribution of DER is offset, yielding solutions that focus on minimizing risk. The value of λ for each subproblem and the total number of subproblems is determined by $\Delta\lambda$, as shown in Equation (4).

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TABLE I: The comparative analysis of averaged results reveals that incorporating EM substantially enhances optimization performance.

Mechanism	without EM	**with EM**	Improvement
HV	0.6562	**0.6728**	2.54%
HV rate	97.35%	**100%**	2.72%
Hit rate	4.98%	**100%**	1,907 %
IGD	4.16E-04	**0.00E+00**	-
GD	3.54E-04	**0.00E+00**	-
NDS#	248.90	**3,844.31**	1,445%

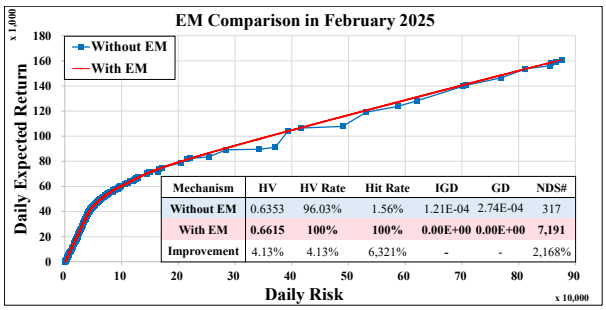

Fig. 1: EM enhances performance by increasing both the diversity and quality of non-dominated solutions.

$$\lambda_k = k \cdot \Delta\lambda \quad (k = 0, 1, 2, \ldots, \frac{1}{\Delta\lambda}) \tag{4}$$

By using a smaller $\Delta\lambda$, the trade-off between risk and return is explored more finely, better capturing investor preferences and revealing a broader range of optimal solutions.

C. Entanglement Move (EM)

After obtaining NDS from the MoQA, the EM mechanism is applied on a classical computer. Inspired by the quantum entanglement phenomenon, the EM leverages the strong correlations between stocks within the portfolio that arise from market dynamics. By systematically adjusting portfolio allocation, the EM guides the search toward potential neighboring solutions around the original NDS, thereby enhancing the overall solution quality. Without the constraints of quantum hardware, the EM is iteratively reapplied until no further improvements are achieved.

III. Experimental Results

In our experiments, the investment targets are the constituents of the Dow Jones Industrial Average (DJIA) index in the U.S. The investment period spans from January 2022 to April 2025, totaling 40 months with monthly re-evaluations.

We first demonstrate the effectiveness of integrating the EM mechanism into MoQA. In this experiment, the value of $\Delta\lambda$ is set to 0.025. The average results are summarized in Table I, with a detailed example illustrated in Fig.1. Comparative indicators such as hit rate exhibit highly significant improvements. As shown in Fig.1, MoQA performs well in low-risk areas, while the EM mechanism significantly enhances performance in high-return regions and expands the diversity of the Pareto front, highlighting the effectiveness of the hybrid method.

Next, we compare the performance of our hybrid method with classical MOO algorithms, PAES [4], SMS-EMOA [5], NSGA-II [6], and MOEA/D [7]. The parameter settings for all algorithms follow those specified in their respective original papers. As shown in Table II, the hybrid approach achieves higher HV, HV rate, and hit rate, alongside lower GD and IGD, indicating superior convergence and diversity. Fig. 2 present zoomed-in views, illustrating that the hybrid approach generates an NDS that extends further toward the top and left, offering portfolio options with higher returns and lower risk compared to classical MOO algorithms. This demonstrates the effectiveness of the hybrid approach in addressing real-world multi-objective portfolio optimization problems.

TABLE II: The comparison between the proposed hybrid method and classical algorithms shows that hybrid achieves higher quality and diversity of solutions.

Algorithm	PAES	SMS-EMOA	NSGA-II	MOEA/D	**Hybrid**
HV	0.2806	0.5266	0.5670	0.6066	**0.6730**
HV rate	40.09%	77.29%	83.70%	89.55%	**99.99%**
Hit rate	0.00%	0.12%	0.04%	1.50%	**98.37%**
IGD	5.66E-03	3.92E-03	2.57E-03	2.13E-03	**1.91E-06**
GD	1.96E-02	5.74E-03	4.39E-03	2.37E-03	**1.51E-06**
NDS#	37.15	94.64	94.12	420.44	**3,844.31**

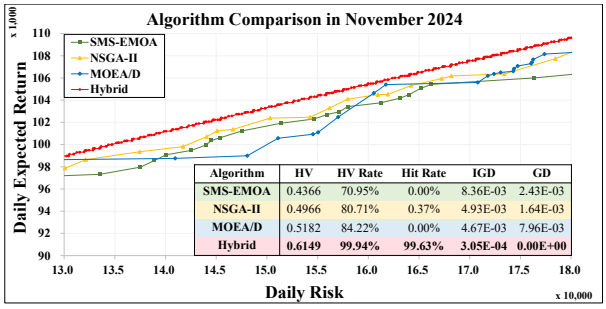

Fig. 2: The hybrid approach delivers better non-dominated solutions than other classical MOO algorithms.

IV. Conclusion

This paper proposes a novel hybrid approach that integrates the advantages of both quantum and classical computing to address multi-objective weighted portfolio optimization. Experimental results demonstrate that the hybrid approach surpasses both MoQA and classical algorithms in terms of convergence and diversity, thereby offering more portfolio options to accommodate various investor preferences. Future work will extend this approach to international cross-market.

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